# Ifeanyi Desmond Dike

Senior Software Engineer | Backend & Voice AI

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## PROFILE

Senior software engineer with **8+ years** in backend and real-time systems. Builds and debugs **production voice AI** using Python, Go, React and TypeScript. Works remotely with US and Canadian teams.

## PROFESSIONAL EXPERIENCE

### DataStrut AI

Founding AI Engineer / Senior Software Engineer | Aug 2025 - Present | New York, US; Remote

- Reduced memory use by **over 70% across two production FastAPI services** through profiling with memray and Gunicorn preloading.
- Migrated production voice AI from Vapi to self-hosted Python/Pipecat, integrating telephony, streaming speech, LLMs and PostgreSQL reporting.
- Cut steady-state speech-recognition sessions from **3 to 1 per call** in three-agent workflows.
- **Diagnosed interruptions and dead air** by tracing stale model decisions and timer races through timestamped voice pipeline events.
- **Excluded speech cancelled before playback release** from recordings; used Twilio dual-channel audio to align both speakers on one timeline.
- Built Fly.io call admission with **atomic, versioned Redis reservations** and active-work checks. Separated scheduling and reporting with dedicated RQ workers, capacity-aware retries and duplicate-call protection.
- Ran voice load tests reaching **40 occupied phone lines**; **48 populated reports (96 jobsites)** were ready within **121.2 seconds** of AI-session end. Identified audio-loss and call-setup failures.
- Built **React report editing** with version checks, history, account isolation and drafts preserved after failed refreshes.

### Fynance Hub

Technical Lead | Oct 2024 - Present | Toronto, Canada; Remote, part-time

- Built a **React/TypeScript mortgage platform** with Go/Python, PostgreSQL and Redis, generating **hundreds of leads** for Canadian brokers.
- Worked with the founder to turn broker requirements into features, including RAG recommendations and outbound voice verification.
- **Separate contract:** building MajorDairy AI's React Native/Expo app and FastAPI backend for dairy-farm workflows. Deployed on **Railway**; app in Google Play testing. Built a DairyComp 305 record importer.

## PROFESSIONAL EXPERIENCE - CONTINUED

### Arbor Platform Inc.

Senior Software Engineer | Feb 2023 - Oct 2024 | New York, US; Remote

- Built **React/TypeScript interfaces** for a content studio, B-roll tools and social publishing; integrated DatoCMS through GraphQL.
- Built Claude/OpenAI video and transcription workflows, reducing production time by **60%** and increasing throughput by **80%**.

### Howdy Learning

Senior Software Development Engineer | May 2020 - Mar 2023 | Remote

- Reduced **WebSocket network traffic by over 99%**; built live-video infrastructure supporting up to **10,000 concurrent connections**.
- Built GetStream/WebSocket chat serving about **100 communities**.
- Delivered video, voice and SSO integrations for Thinkific and WordPress, used by **over 100 customers**.

### Bujuwa Nigeria Limited

Software Developer | Jul 2018 - Dec 2019 | Abuja, Nigeria

- Built web and mobile assessment systems, from requirements through design and delivery.

## SELECTED INDEPENDENT PROJECT

**Cenphi** | Co-Founder and Lead Developer | Mar 2025 - Present

- Built **React/TypeScript and Go/Python video workflows** with SSE progress updates. Added connection cleanup, completion deduplication, session isolation, access checks and regression tests.

## TECHNICAL SKILLS

**Backend:** Python, asyncio, Go, FastAPI, Flask, gRPC, PostgreSQL, Redis, MongoDB, RQ.

**Frontend:** React, TypeScript, JavaScript, GraphQL (DatoCMS), React Native, Expo.

**Infrastructure:** Fly.io, Heroku, Railway, Docker, CI/CD, observability; additional exposure to AWS and Kubernetes.

**Voice & AI:** Pipecat, Twilio, WebSockets, WebRTC, streaming STT/TTS, VAD, LLM tools, RAG, PyTorch, Hugging Face.

## RESEARCH, OPEN SOURCE & WRITING

**Voice-agent evaluation:** Sole-author preprints on ATR and Full-Duplex-Bench reproduction, with ATR code and the benchmark code. Four co-authored IEEE IEMCON papers (2018).

**Pipecat:** Reported an Anthropic integration bug (#5035 and #5036); maintainer fix: #5041.

**Technical writing:** Debugging voice recordings; profiling Python memory in production.

## EDUCATION

**M.Sc., Financial Engineering** | WorldQuant University | Dec 2025 - Dec 2027 (expected; in progress)

**B.Tech., Applied Mathematics, First Class** | Federal University of Technology, Akure | 2018
GPA: 4.63/5.00. National university mathematics competition: Gold (2017), Bronze (2016).